

MSE 641 - Text Analytics: Project Guide

Spring 2026

Project Overview

The course project constitutes 70% of your final grade in MSE 641. It provides an opportunity to apply your text analytics skills to practical tasks. You have two options:

1. **Default Project:** Clickbait Detection Challenge
2. **Custom Project:** Design your own NLP project (individually or in teams of 2)

Teams of 2 are encouraged but will be expected to complete more substantial projects than individuals.

Project Timeline

Deliverable	Due Date	Weight
Project Proposal (Custom only)	May 27, 2026 at 11:59 PM	No grade
Project Milestone	June 24, 2026 at 11:59 PM	10% of course grade
Final Project Report	August 5, 2026 at 11:59 PM	60% of course grade

All students (Due: May 23)

- Fill out the Survey "Project teams" on Learn, specifying the type of project you will be working on and your team membership.

Option 1: Default Project - Clickbait Detection Challenge

The default project involves participating in the Clickbait Detection Challenge with two specific tasks:

- **Task 1:** [Kaggle Competition Link](#)

- **Task 2:** [Kaggle Competition Link](#)

Default Project Requirements:

For Project Proposal (Due: May 27)

- No Learn submission required.

For Project Milestone (Due: June 24)

- Train a baseline model and submit it to both Kaggle leaderboards.
- You must have at least one successful submission for both Task 1 and Task 2.
- The simplest approach is to run the baselines included in the competition description.
- Ensure output files are in CSV format.

For Final Project (Due: August 5)

- Continue improving your models and making submissions to the leaderboard.
- Prepare a final report following the guidelines in the "Final Report Guidelines" section.

Option 2: Custom Project

For the custom project, you can design your own NLP task individually or in a team of 2 students.

Custom Project Ideas:

- Conversational (dialogue) models with machine responses conditioned on a specific persona/character
- Neural network models for product recommendation based on customer reviews
- Story generation
- Multimodal NLP applications

Recommended Datasets:

- Amazon product reviews: <http://jmcauley.ucsd.edu/data/amazon/>
- OpenSubtitles multilingual movie dialogue corpus: <http://opus.nlpl.eu/OpenSubtitles.php>
- Cornell Movie dialogue corpus: http://www.cs.cornell.edu/~cristian/Cornell_Movie-Dialogs_Corpus.html
- Quora question pairs: <https://www.kaggle.com/c/quora-question-pairs>

- Toxic comment classification: <https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge>
- Yelp reviews: <https://www.yelp.com/dataset/challenge>

Custom Project Requirements:

For Project Proposal (Due: May 27)

Submit a one-page proposal to Learn Drop Box "Project Proposal" containing:

1. **Team:** Names and student IDs of all team members
2. **Problem Description** (1-2 sentences): What problem will you investigate?
3. **Data** (1-3 sentences): What dataset will you use? How will you collect it?
4. **Methodology/Algorithm** (2-4 sentences): What method or algorithm are you proposing? Will you use existing implementations? How will you modify them?
5. **Related Work** (3+ prior works): Which papers will inform your understanding?
6. **Evaluation Plan** (2-3 sentences): How will you evaluate results? What metrics will you use?
7. **Minimal Requirements** confirmation:
 - Dataset with at least 10,000 labeled examples for supervised methods
 - Ability to collect complete dataset by milestone due date
 - Task feasibility (prior work exists or humans can achieve good accuracy)
 - Identified automatic evaluation metrics
 - Justification if no automatic evaluation is possible
 - NLP is required for good performance on the task

Also complete the Learn Survey Form "Project teams" with your team members.

For Project Milestone (Due: June 24)

Submit a one-page report to Learn Drop Box "Project Milestone" containing:

1. **Team:** Names and student IDs
2. **Problem Description:** What problem are you investigating?
3. **Data:** What dataset are you using? How did you collect it?
4. **Baseline:** What is your baseline algorithm? (Can be simple: bag-of-vectors, Naive Bayes, n-gram language model)
5. **Evaluation Methodology:** How are you evaluating your baseline?
6. **Results:** How well does your baseline work?

By this point, you must have:

- Collected all your data
- Implemented a simple baseline
- Set up your evaluation pipeline
- Run your baseline on your data and evaluated its performance

For Final Project (Due: August 5)

Follow the "Final Report Guidelines" section below.

Final Report Guidelines

Your final report should be 6-8 pages using the [ACL template](#).

Submission Requirements:

Submit to Learn Drop Box "Final Project Report":

1. **Final Project Report:** PDF file in ACL format
2. **Supplementary Materials:** Zip file containing all project code
3. **GitHub Repository Link:** Include in your report
4. **Final Project Additional Information:** PDF file with the details outlined below

Only one submission per team is required, but ensure all team members are tagged.

Report Structure:

1. **Title and Author(s)**
2. **Abstract** (max 300 words)
3. **Introduction:** Introduce the problem and your approach
4. **Background/Related Work:** Discuss relevant literature
5. **Approach:** Detail your methodology (include equations, figures, plots as needed)
6. **Experiments:**
 - Dataset(s) used
 - Experimental setup (model configurations, learning rate, training time, etc.)
 - Evaluation metric(s)
 - Results (both quantitative and qualitative evaluation)
7. **Conclusion:** What you learned and future ideas
8. **References:** All literature that informed your work

Supplementary Materials:

Include:

- Source code (required)
- Optional: videos, interactive visualizations, demos

Do not include:

- Entire submodules (e.g., Theano, Caffe, CoreNLP)
- Code larger than 1MB
- Model checkpoints

Final Project Additional Information

Submit one PDF per team containing the following information with your final report:

All Teams Must Include:

- **Contribution per team member:** For teams of 2, detail what each person contributed to the project (1-2 sentences per person)

Default Project Teams Must Also Include:

1. **Kaggle username** of the team member who has consistently made all the submissions
2. **Kaggle submission details** for your best test leaderboard submission (which will contribute to your grade):
 - 'weightedScore' as it appears on the leaderboard
 - 'Date of Entry' of submission as it appears on the leaderboard
 - Submitter as it appears on the leaderboard
 - Name of the zip file corresponding to your best submission

Academic Integrity

In order to maintain a culture of academic integrity, members of the University of Waterloo community are expected to promote honesty, trust, fairness, respect, and responsibility. Check the [Office of Academic Integrity](#) for more information.

For grievances, discipline issues, or appeals, refer to University Policies:

- Policy 70 (Student Petitions and Grievances)
- Policy 71 (Student Discipline)
- Policy 72 (Student Appeals)

Text matching software (Turnitin®) may be used to screen project deliverables and exams.

Support for Students with Disabilities

AccessAbility Services, located in Needles Hall, room 1132, collaborates with academic departments to arrange appropriate accommodations for students with disabilities without compromising academic integrity. If you require accommodations, register with AccessAbility Services at the beginning of each academic term.